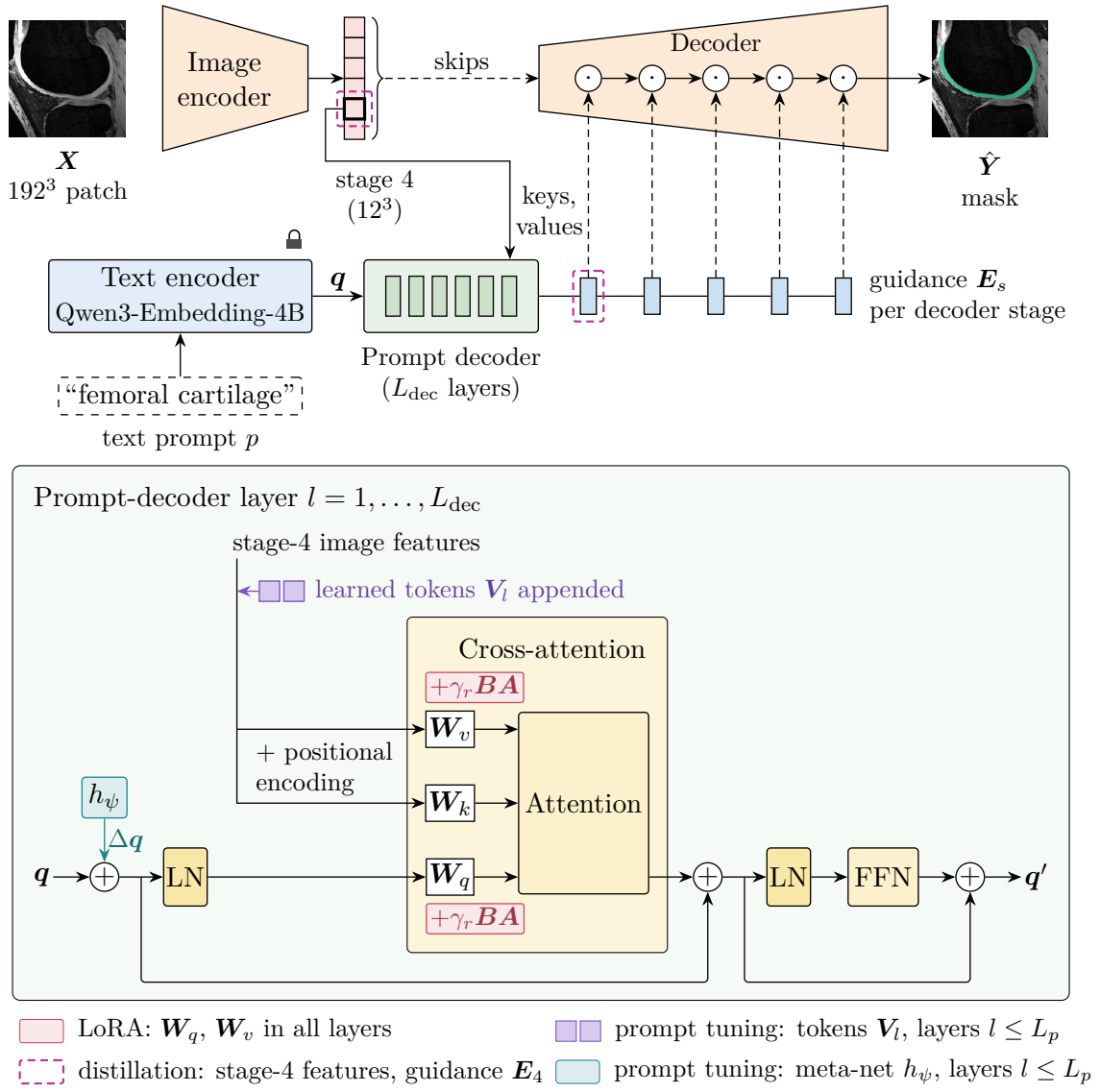

Image encoder
skips
Decoder
$\boldsymbol{X}$
192^3 patch
stage 4
(12^3)
keys, values
$\hat{\boldsymbol{Y}}$
mask
Text encoder
Qwen3-Embedding-4B
$\boldsymbol{q}$
Prompt decoder
(L_{dec} layers)
guidance $\boldsymbol{E}_s$
per decoder stage
"femoral cartilage"
text prompt p
Prompt-decoder layer $l = 1, \ldots, L_{\text{dec}}$
stage-4 image features
learned tokens $\boldsymbol{V}_l$ appended
Cross-attention
$+\gamma_r \boldsymbol{B}\boldsymbol{A}$
$\boldsymbol{W}_v$
$\boldsymbol{W}_k$
$\boldsymbol{W}_q$
Attention
+ positional encoding
h_ψ
$\Delta \boldsymbol{q}$
$\boldsymbol{q}$
LN
LN
FFN
$\boldsymbol{q}'$
LoRA: $\boldsymbol{W}_q$, $\boldsymbol{W}_v$ in all layers
prompt tuning: tokens $\boldsymbol{V}_l$, layers $l \leq L_p$
distillation: stage-4 features, guidance $\boldsymbol{E}_4$
prompt tuning: meta-net h_ψ, layers $l \leq L_p$